MH1300 FOUNDATIONS OF MATHEMATICS

Tutorial 4 Solutions

Ex. 3.1.8. Let $B(x)$ be "$-10 < x < 10$". Find the truth set of $B(x)$ for each of the following domains:

(a) $\mathbb{Z}$ (b) $\mathbb{Z}^+ = \{n \in \mathbb{Z} \mid n > 0\}$ (c) The set of even integers.

SOLUTION . (a) Truth set is $\{x \in \mathbb{Z} \mid B(x)\} = \{x \in \mathbb{Z} \mid -10 < x < 10\} = \{-9, -8, -7, -6, \cdots, 6, 7, 8, 9\}$.

(b) Truth set is $\{x \in \mathbb{Z}^+ \mid B(x)\} = \{x \in \mathbb{Z}^+ \mid -10 < x < 10\} = \{1, 2, 3, 4, 5, 6, 7, 8, 9\}$.

(c) Truth set is $\{x \in \text{set of even integers} \mid B(x)\} = \{x \in \text{set of even integers} \mid -10 < x < 10\} = \{-8, -6, -4, -2, 0, 2, 4, 6, 8\}$. Remember that even integers include negative even numbers like $-2, -4, -6, \cdots$ as well as 0. □

Ex. 3.1.12. Find counterexamples to show that the statement is false.

$$\forall \text{ real numbers } x \text{ and } y,\ \sqrt{x+y} = \sqrt{x} + \sqrt{y}.$$

SOLUTION . Counterexample 1: Let $x = 1$ and $y = 1$, and note that

$$\sqrt{1+1} = \sqrt{2} \neq 2 = 1 + 1 = \sqrt{1} + \sqrt{1}.$$

(Any real numbers x and y with $xy \neq 0$ will produce a counterexample.)

Counterexample 2: Let $a = 1, b = -1$. Then $\sqrt{a+b} = \sqrt{0} = 0$, is a real number. However, $\sqrt{a} + \sqrt{b} = \sqrt{1} + \sqrt{-1}$ is not a real number. □

Ex. 3.1.30. Rewrite each statement without using quantifiers or variables. Indicate which are true and which are false, and justify your answers as best as you can.

Let the domain of x be the set $\mathbb{Z}$ of integers, and let $\text{Odd}(x)$ be "x is odd", $\text{Prime}(x)$ be "x is prime", and $\text{Square}(x)$ be "x is a perfect square". (An integer is a **perfect square** if and only if it equals the square of some integer. For example, 25 is a perfect square because $25 = 5^2$.)

a. $\exists x$ such that $\text{Prime}(x) \wedge \neg\text{Odd}(x)$.
b. $\forall x$, $\text{Prime}(x) \to \neg\text{Square}(x)$.
c. $\exists x$ such that $\text{Odd}(x) \wedge \text{Square}(x)$.

SOLUTION . a. This statement translates as "There is a prime number that is not odd." This is true. The number 2 is prime and it is not odd.

b. This statement translates as "If an integer is a prime number number, then it is not a perfect square." This is true.

Let n be a prime number. Then $n > 1$ and n is not a product of two smaller positive integers. So it cannot be a perfect square.

c. This statement translates as "There is a number that is both an odd number and a perfect square." This is true. For example, the number 9 is odd and it is also a perfect square (because $9 = 3^2$).

$\square$

Ex. 3.2.15. Let $D = \{-48, -14, -8, 0, 1, 3, 16, 23, 26, 32, 36\}$. Determine which of the following statements are true and false, and justify your answer.

(b) $\forall x \in D$, if x is less than 0 then x is even.
(d) $\forall x \in D$, if the ones digit of x is 2, then the tens digit of x is 4 or 5.

SOLUTION . (b) This is a universal (conditional) statement, so to show that it is true, we shall need to test and exhaustively verify the predicate for every element $x \in D$:

$x = -48$: "if -48 is less than 0 then -48 is even" is TRUE, since both the premise "$-48 < 0$" and the conclusion "-48 is even" are true.
$x = -14$: "if -14 is less than 0 then -14 is even" is TRUE, since both the premise and the conclusion are true.
$x = -8$: "if -8 is less than 0 then -8 is even" is TRUE, since both the premise and the conclusion are true.
$x = 0, 1, 3, 16, 23, 26, 32, 36$: "if x is less than 0 then x is even" is TRUE, since the premise $x < 0$ is false.

So by the method of exhaustion, the statement "$\forall x \in D$, if x is less than 0 then x is even" is TRUE.

Alternatively we can use the simplification rule given in the lecture, since this is a universal conditional statement, we note that

"$\forall x \in D$, if x is less than 0 then x is even"

is logically equivalent to

"$\forall x \in \{y \in D \mid y < 0\}$, x is even"

So we verify the second statement by the method of exhaustion. Now we need to test every element in the new domain, $\forall x \in \{y \in D \mid y < 0\} = \{-48, -14, -8\}$. It is easy to see that for each $x = -48, -14$ and -8, the x is even. So the second statement is true.

(d) Let's use the simplification rule. Now the new domain is $\{y \in D \mid$ the ones digit of y is $2\} = \{32\}$. So we need to check that for $x = 32$, the tens digit of x is 4 or 5. This is of course false, so the statement is FALSE.

Alternatively, if we don't use the simplification rule, we shall need to show directly that the statement

"$\forall x \in D$, if the ones digit of x is 2, then the tens digit of x is 4 or 5"

is FALSE. To show a universal (conditional) statement is false, we shall need to find (at least) one counterexample. Take $x = 32 \in D$. Then we shall check that $x = 32$ provides the counter-example, by verifying that the predicate fails on $x = 32$. The predicate is

"if the ones digit of x is 2, then the tens digit of x is 4 or 5"

so when substituting in $x = 32$, it becomes

"if the ones digit of 32 is 2, then the tens digit of 32 is 4 or 5" which is obviously false since the premise is true but the conclusion is false. So the universal (conditional) statement is false by the counter-example $x = 32$. $\square$

Ex. 3.2.17, 27. Write a negation, the contrapositive, converse, and inverse of the following statement.

$$\forall \text{ integers } d, \text{ if } 6/d \text{ is an integer then } d = 3.$$

SOLUTION . **Negation:** $\exists$ an integer d such that $6/d$ is an integer and $d \neq 3$.

Contrapositive: $\forall$ integers d, if $d \neq 3$ then $6/d$ is not an integer.

Converse: $\forall$ integers d, if $d = 3$ then $6/d$ is an integer.

Inverse: $\forall$ integers d, if $6/d$ is not an integer then $d \neq 3$. $\square$

Ex. 3.2.47. Use the facts that the negation of a $\forall$ statement is a $\exists$ statement and that the negation of an if-then statement is an *and* statement to rewrite the following statement without using the words *necessary* and *sufficient*.

The absence of error messages during translation of a computer program is only a necessary and not a sufficient condition for reasonable [program] correctness.

SOLUTION . **Formal Version:** $\forall$ computer programs P, if P is correct then P translates without error messages. However, $\exists$ a computer program P such that P translates without error messages and P is not correct.

Informal Version: If a computer program is correct, then it translates without error messages. But some incorrect computer programs also do not produce error messages during translation. $\square$

Ex. 3.3.12. Let $D = E = \{-2, -1, 0, 1, 2\}$. Write negations for each of the following statements and determine which is true, the given statement or its negation.

a. $\forall x \in D$, $\exists y \in E$ such that $x + y = 1$.
b. $\exists x \in D$ such that $\forall y \in E, x + y = -y$.

SOLUTION . a. $\exists x \in D$ such that $\forall y \in E$, $x + y \neq 1$.

The negation is true. Take $x = -2$. Then for any y in E, $x + y < 1$.

b. $\forall x \in D$, $\exists y \in E$ such that $x + y \neq -y$.

The negation is true. No matter what number x is given, we may choose a y so that $x + y \neq y$.

Given x	Choose y	$x + y$	$-y$
-2	0	-2	0
-1	0	-1	0
0	1	1	-1
1	0	1	0
2	0	2	0

□

Ex. 3.3.35, 36. (a) Rewrite the statement formally using quantifiers and variables, and (b) write a negation for the statement.

35. Everybody trusts somebody.
36. Somebody trusts everybody.

SOLUTION . **35.** (a): $\forall$ people x, $\exists$ a person y such that x trusts y.

(b): Negation: $\exists$ a person x such that $\forall$ people y, x does not trust y.
Or: Somebody trusts nobody.

36. (a): $\exists$ a person x such that $\forall$ people y, x trusts y.

(b): Negation: $\forall$ people x, $\exists$ a person y such that x does not trust y.
Or: Nobody trusts everybody. □

Ex. 3.3.58. Let $P(x)$ and $Q(x)$ be predicates and suppose D is the domain of x. For the statement forms in each pair, determine whether (a) they have the same truth value for every choice of $P(x)$, $Q(x)$, and D, or (b) there is a choice of $P(x), Q(x)$, and D for which they have opposite truth values.

$$\exists x \in D, (P(x) \wedge Q(x)), \qquad \text{and} \qquad (\exists x \in D, P(x)) \wedge (\exists x \in D, Q(x))).$$

SOLUTION . They do not necessarily have the same truth values.

Take $D = \mathbb{R}$, and let $P(x)$ be "x is positive" and let $Q(x)$ be "x is negative".

Then "$\exists x \in D, (P(x) \wedge Q(x))$" can be written "$\exists$ a real number x such that x is both positive and negative," which is false.

On the other hand, "$(\exists x \in D, P(x)) \wedge (\exists x \in D, Q(x))$" can be written "$\exists$ a real number that is positive and $\exists$ a real number that is negative", which is true.

Other possible choices:
$D = \mathbb{R}$ and $P(x)$ be "$x = 1$" and $Q(x)$ be "$x \neq 1$".
$D = \mathbb{Z}$ and $P(x)$ be "x is odd" and $Q(x)$ be "x is even".
Or you can have your own answer! □